MIDTERM PROJECT: PYTHON MANAGEMENT SYSTEM

Team Size: 3 students
Project Weight: 13 points
Final Submission: October 18, 2026, 23:59
Project Defense: October 19–21, 2026, during regular class hours

Project Description

Students will work in teams of **3** to develop a **console-based Python Management System**.

Each team must select **one** of the six project topics below.

The project must demonstrate the use of:

- conditions and loops;
- functions;
- input/output;
- Object-Oriented Programming (OOP);
- file handling;
- SQLite database.

The final program must be functional and executable.

Project Topics

No.	Project	Main Data
1	**Hotel Reservation System**	Rooms, Guests, Reservations
2	**Airline Reservation System**	Flights, Passengers, Tickets
3	**Hospital Management System**	Patients, Doctors, Appointments
4	**Library Management System**	Books, Members, Borrowings
5	**Online Store Management System**	Products, Customers, Orders
6	**Car Rental Management System**	Cars, Customers, Rentals

Tasks to Complete

1. Add Information

The program must allow the user to add and save new records.

Project	Required Records
Hotel	Room, Guest, Reservation
Airline	Flight, Passenger, Ticket
Hospital	Patient, Doctor, Appointment
Library	Book, Member, Borrowing
Online Store	Product, Customer, Order
Car Rental	Car, Customer, Rental

All records must be stored in an **SQLite database** with at least **3 related tables**.

2. Display Records

The program must display stored records in a clear and readable format.

Students must be able to view the main records related to their selected project.

For example, the Hotel system should display rooms, guests, and reservations; the Library system should display books, members, and borrowings.

3. Calculate Results and Statistics

The program must perform at least **two calculations or statistical operations**.

Project	Required Calculations
Hotel	Total revenue; occupancy rate
Airline	Total ticket revenue; seat occupancy rate
Hospital	Total appointments; appointments per doctor
Library	Number of borrowed books; most frequently borrowed book
Online Store	Total sales revenue; best-selling product
Car Rental	Total rental revenue; most frequently rented car

Calculations must use data stored in the database.

4. Search for Information

The program must provide at least **two search or filtering options**.

Project	Search / Filter Examples
Hotel	Search guest by name/ID; show available rooms
Airline	Search passenger by name/ID; search flights by destination
Hospital	Search patient by name/ID; show appointments by doctor
Library	Search book by title; search member by name/ID
Online Store	Search product by name; search customer by name/ID
Car Rental	Search car by model; search customer by name/ID

If no record is found, the program must display an appropriate message.

5. Update and Delete Information

The program must allow the user to:

- update an existing record;
- delete or cancel an existing record.

The project must demonstrate the basic **CRUD operations: Create, Read, Update, Delete**.

6. Save and Export Results

The program must generate a summary report and save it as:

`results.txt` or **`results.csv`**

The exported file must include the main project information, calculations, and statistics.

Dataset Guidelines

Each team must prepare the following minimum dataset:

Project	Minimum Dataset
Hotel Reservation System	50 rooms, 30 guests, 30 reservations
Airline Reservation System	20 flights, 50 passengers, 40 tickets
Hospital Management System	50 patients, 10 doctors, 40 appointments
Library Management System	50 books, 30 members, 30 borrowings
Online Store Management System	50 products, 30 customers, 40 orders
Car Rental Management System	30 cars, 30 customers, 30 rentals

Data may be entered manually or generated programmatically.

The required dataset must be available in the submitted database.

Programming Requirements

Each project must include:

- **Functions** for the main program operations;
- at least **2 meaningful classes**;
- **SQLite** database with at least 3 related tables;
- **file handling** for exporting results;
- appropriate **input validation**.

Each team must also implement at least **TWO** of the following:

- Regular Expressions (`re`);
- Random (`random`);
- Inheritance;
- Method Overriding;
- Data Hiding / Encapsulation.

The selected features must be used meaningfully in the program.

Console Menu

The application must contain a clear menu adapted to the selected project.

Example:

```
MANAGEMENT SYSTEM

1. Add Information
2. Display Records
3. Search
4. Update
5. Delete
6. Show Statistics
7. Export Results
0. Exit

Choose an option:
```

The program must continue running until the user selects **Exit**.

Evaluation Criteria — 13 Points

Criterion	Points
Functionality and logic	**4**
Use of OOP, Files, Database	**3**
Code clarity and correctness	**2**
Report and presentation	**2**
Teamwork and individual defense	**2**
TOTAL	**13**

Submission

Submit all project files as **one ZIP archive**.

File name: `GroupID_ProjectName.zip`

Example: `Group04_HotelReservation.zip`

The archive must contain:

`GroupID_ProjectName.zip`

```
├── main.py
├── project.db
├── results.txt / results.csv
├── ProjectReport.pdf
└── ProjectPresentation.pptx
```

The **Project Report** must be **3–5 pages** and include:

- project title and team members;
- brief project description;
- database structure;
- main functions and classes;
- screenshots of the working program;
- calculations and results;
- contribution of each team member.

A **PowerPoint presentation is mandatory for the project defense**.

The presentation should contain approximately **5–7 slides** covering the project objective, database structure, main functionality, program results, and team member contributions.

Submit the complete ZIP archive through **TEAMS Assignment**.

Project Defense

The project defense will take place **October 19–21, 2026, during regular class hours**.

Each team must:

- give a presentation of the project;
- demonstrate the working program;
- demonstrate the main CRUD operations;
- demonstrate search and calculations;
- explain the database structure;
- explain the main functions and classes;
- explain the two selected additional features;

Every team member must participate in the presentation and defense.

The instructor may ask any student to explain a part of the code or make a small modification to the program during the defense.

Individual grades may differ depending on each student's contribution and performance during the defense.

Deadlines

Task	Deadline
Final Submission	**October 18, 2026, 23:59**
Project Defense	**October 19–21, 2026, during regular class hours**

Late submissions will not be accepted!